Chapter 2 — Probability sampling

Collection instruments gather answers; sampling decides whose answers count for inference

Learning objectives

- Explain why sampling underpins generalization, describe simple random, stratified
- Cluster probability designs
- Match each design to a typical population and logistics profile

Why sampling matters

- Sampling lets researchers learn about a population from a manageable subset of cases
- A well-chosen sample supports generalization
- This part covers three probability designs that appear repeatedly in practice
- Each assumes a usable sampling frame unless noted otherwise

Simple random sampling

- In simple random sampling, each population member has an equal chance of selection
- The method reduces selection bias when a complete frame exists and when the sample is
- Operationally, teams build a complete list, draw units with a random process
- Its practical limits are incomplete frames and non-response

Example 2.10 — Simple random draw from an employee roster

- Example 2.10 — hands-on module
- Example 2.10 shows a clean frame case
- A firm with two thousand employees on a complete human-resources roster wants feedback on
- Analysts draw two hundred employee identifiers uniformly at random and invite those people
- If response is high
- View files: `modules/chapter2/example10/`

Stratified sampling

- Stratified sampling first partitions the population into meaningful subgroups
- The design improves precision when subgroups differ on the outcome of interest and
- Costs include planning complexity and the need for prior knowledge to define strata

Example 2.11 — Stratified sample of cart abandoners

- Example 2.11 — hands-on module
- Example 2.11 mirrors the retail cart-abandonment study from the chapter opener
- Analysts divide recent abandoners into mobile and desktop strata
- The design ensures that device-specific checkout friction is visible even if one device
- Explore the chapter example module
- View files: `modules/chapter2/example11/`

Cluster sampling

- Cluster sampling selects groups of units, schools, clinics, city blocks
- It is attractive when individual frames are unavailable or when travel costs make visiting
- Clusters can be efficient
- Prefer clusters that still reflect population diversity when logistics dominate

Takeaways

- Use random sampling when the population is relatively homogeneous and a complete list
- Use stratified sampling when key subgroups must appear in sufficient numbers
- Use cluster sampling when logistics dominate and entire sites can be visited as units
- Non-probability options for weak frames appear in the next part

Next

- Complete the quiz for this part
- Complete the quiz